

Dot & Key — Brand Health & Customer Sentiment Analysis

A Data Analytics Case Study

Built by Kanishka Kumari | Data Analyst | Power BI · Python · SQL

Project Overview

This project was built as an unsolicited data initiative to demonstrate analytical capability to Dot & Key — one of India's fastest-growing D2C skincare brands. The goal: scrape real customer reviews, analyse sentiment, identify key pain points, and present actionable business insights through an interactive Power BI dashboard — all in a single day.

Tools & Technologies

Tool	Purpose
Python + Google Colab	Web scraping, data cleaning, sentiment tagging
google-play-scraper	Scraped 1,000 real reviews from Google Play Store
Pandas	Data transformation, keyword frequency analysis, CSV export
Power BI Desktop	Interactive 4-page dashboard with navigation buttons
DAX	Custom measures: Sentiment %, Reply Rate %, Negative Replied

Data Collection & Methodology

Data was scraped from Dot & Key's official Google Play Store app (ID: com.dotandkey) using the google-play-scraper Python library. Initial attempts to scrape Flipkart and the brand's own website were blocked by bot protection — the Play Store provided cleaner, more reliable data covering Aug 2025 – Jun 2026.

- 1,000 customer reviews sorted by newest first
- Star ratings (1–5 scale), full review text, reviewer name, date
- Brand reply content and reply timestamps (used for response rate analysis)

Data Processing & Feature Engineering

Sentiment Classification: Positive (4–5 stars), Neutral (3 stars), Negative (1–2 stars) applied via Python function.

Theme Tagging: Reviews tagged across 4 themes using keyword matching: Delivery & Orders, App Experience, Customer Service, Product Quality.

Keyword Frequency Analysis: Top 20 keywords extracted from positive and negative reviews separately after stopword removal.

Month Column: YYYY-MM format derived from date field for time-series trend analysis in Power BI.

Dashboard Structure — 4 Pages

Page 1 — Executive Summary: KPI cards (Total Reviews, Positive %, Negative %, Neutral %), monthly sentiment trend line chart (Sep 2025–Jun 2026), sentiment donut chart, page navigation buttons.

Page 2 — What Are Customers Complaining About?: Horizontal bar chart of Top Customer Pain Points across 4 themes. Delivery & Orders leads at 395 mentions — nearly 2x the next category. Key insight text box included.

Page 3 — How Well Does Dot & Key Listen?: KPI cards: Total Replied (656), Reply Rate % (65.6%), Negative Replied (327), Positive Replied (308). Shows 111 negative reviews went unanswered.

Page 4 — Customer Complaints: In Their Own Words: Table of top critical negative reviews with full text, year, month, sentiment. Real voices making data human and actionable for brand team.

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Key Findings & Business Insights

Key Metrics at a Glance

Metric	Value	What It Means
Total Reviews Analysed	1,000	Full dataset — no sampling bias
Positive Sentiment	53.0%	Majority of customers are happy with products
Negative Sentiment	43.8%	Alarmingly high — needs urgent attention
Neutral Sentiment	3.2%	Very few fence-sitters — strong opinions either way
Brand Reply Rate	65.6%	Active engagement but not comprehensive
Negative Reviews Unanswered	111 of 438	Gap in customer service coverage
Top Pain Point	Delivery & Orders	395 keyword mentions — 2x more than any other theme
2nd Biggest Issue	App Experience	217 mentions — login, tracking, UI bugs
Positive Keywords	good, love, amazing, sunscreen	Product quality is genuinely appreciated

Actionable Business Insights

- *Products are loved — 'good', 'love', 'amazing', 'sunscreen', 'skin results' dominate positive reviews. The formulations work.*
- *Customers are being lost at the delivery stage — 395 negative mentions around orders, tracking, and missing products.*
- *App experience is the 2nd biggest issue — login failures, order tracking bugs, and UI issues are hurting retention and repeat purchases.*
- *The BOGO offer in Aug 2025 created a complaints spike — multiple customers received 1 product instead of 2, with no resolution from support.*
- *65.6% reply rate shows active community management — but 111 negative reviews went unanswered, a gap worth closing to protect brand trust.*
- *Recent data (Jun 2026) shows improving positive sentiment (123 positive vs 34 negative) — the brand appears to be course-correcting.*
- *Overall: Dot & Key has a strong product — but risks losing loyal customers due to operational failures in delivery and app experience.*

The Opportunity

If Dot & Key fixes delivery reliability and app experience, they can convert their 43.8% negative reviewers into loyal customers — the product quality is already there to retain them.

This is the kind of insight a dedicated Data Analyst can surface regularly — from customer reviews, sales data, app analytics, and campaign performance — enabling faster, smarter business decisions.

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